

How to do?

Follow the below Steps

After Logging in..., to your tellnet with user ID and password type the commands as below

Stp1: `cat>p1.sh`

//After clicking enter of the above command start typing your 1st program as it is, make sure no errors type very carefully!!!...(p1.sh file is considered as your 1st OS program for 2nd program p2.sh and so onn for all it follows

Step 2: After typing your program press Ctrl+c(which is located on keyboard)

Step 3: `sh p1.sh` (first we created .sh file named p1 it should be same!!!)

And after clicking enter your bash script will be executed if no errors found if error found type the content once again

These steps only should be followed for all the programs, only p1.sh, p2.sh and so on... Names will be changing according to programs

1. Write a shell script to access information using date, history, man, who, whoami, finger, cal.

```
#!/bin/bash
```

```
echo "=== Current Date and Time ==="
```

```
date
```

```
echo "=== Calender ==="
```

```
cal
```

```
echo "\n=== Man Page Preview for 'who' ==="
```

```
man who | head -n 10
```

```
echo "\n=== Logged In Users ==="
```

```
who
```

```
echo "\n=== Current User ==="
```

```
whoami
```

```
echo "\n=== Finger Info for Current User ==="
```

```
finger $(whoami)
```

2. Write a shell script to display contents of files using cat, vi, more, head, tail, grep, cmp, wc.

```
FILE="sample.txt"
```

```
# Display the contents of the file using cat
```

```
echo "Contents of the file using cat:"
```

```
cat $FILE
```

```
# Display the contents of the file using more (page-by-page view)
```

```
echo "Contents of the file using more (press 'q' to quit):"
```

```
more $FILE
```

```
echo "-----"
```

```
# Display the first 10 lines of the file using head
```

```
echo "First 10 lines of the file using head:"
```

```
head $FILE
```

```
echo "-----"
```

```
# Display the last 10 lines of the file using tail
```

```
echo "Last 10 lines of the file using tail:"
```

```
tail $FILE
```

```
echo "-----"
```

```
# Search for a specific word (e.g., 'error') in the file using grep
```

```
echo "Lines containing 'error' using grep:"
```

```
grep "error" $FILE
```

```
echo "-----"
```

```
# Compare two files using cmp
```

```
echo "Comparing the contents of the file with another file (file2.txt):"
```

```
cmp $FILE file2.txt
```

```
echo "-----"
```

3. Write a shell script to manage files & directories using cat, cp, ls, mv, rm, chmod, cd, mkdir, rmdir, mv, pwd.

```
while true; do
```

```
    echo "1. List"
```

```
    echo "2. Make dir"
```

```
    echo "3. Remove dir"
```

```
    echo "4. Where am I"
```

```
    echo "5. Exit"
```

```
read -p "Choose: " c
```

```
case $c in
```

```
1) ls ;;
```

```
2) read -p "Name: " d; mkdir "$d" ;;
```

```
3) read -p "Name: " d; rmdir "$d" ;;
```

```
4) pwd ;;
```

```
5) break ;;
```

```
*) echo "Wrong!" ;;
```

```
esac
```

```
echo
```

```
done
```

1. Write a shell script to implement process utilities using ps, pid, ppid, tty, time, kill, exit

```
while true; do
```

```
echo "1. Show all processes"
```

```
echo "2. Show my PID"
```

```
echo "3. Show my PPID"
```

```
echo "4. Show my terminal (TTY)"
```

```
echo "5. Show time used by my shell"
```

```
echo "6. Kill a process"
```

```
echo "7. Exit"
```

```
read -p "Choose: " ch
```

```
case $ch in
```

```

1) ps ;;
2) echo "PID: $$" ;;
3) echo "PPID: $PPID" ;;
4) tty ;;
5) ps -p $$ -o time ;;
6) read -p "Enter PID to kill: " pid; kill $pid ;;
7) echo "Exiting..."; exit ;;
*) echo "Invalid!" ;;

esac

echo

done

5. Files with read, write, and execute permissions for user in current directory
echo "Files with read, write, and execute permissions for user in current directory:"
echo "-----"

for file in *; do
    if [ -f "$file" ] && [ -r "$file" ] && [ -w "$file" ] && [ -x "$file" ]; then
        echo "$file"
    fi
done

```

Remaining scripts will be updated shortly!!!

<https://github.com/issu321/Study-Material/tree/main>